

Excerpts from Formalized Music

Iannis Xenakis

Chapter V: Free Stochastic Music by Computer

After this interlude, we return to the treatment of composition by machines.

The theory put forward by *Achorripsis* had to wait four years before being realized mechanically. This realization occurred thanks to M. François Génuys of IBM-France and to M. Jacques Barraud of the Régie Autonome des Transports Parisiens.

The Paradox: Music and Computers

A Stochastic Work Executed by the IBM-7090

The general public has a number of different reactions when faced by the alliance of the machine with artistic creation. They fall into three categories:

“It is impossible to obtain a *work of art*, since by definition it is a handicraft and requires moment-by-moment “creation” for each detail and for the entire structure, while a machine is an inert thing and cannot invent.”

“Yes, one may play games with a machine or use it for speculative purposes, but the result will not be “finished”: it will represent only an experiment—interesting, perhaps, but no more.”

The enthusiasts who at the outset accept without flinching the whole frantic brouhaha of science fiction. “The moon? Well, yes, it’s within our reach. Prolonged life will also be with us tomorrow—why not a creative machine?” These people are among the credulous who, in their, idiosyncratic optimism, have replaced the myths of Icarus and the fairies, which have decayed, by the scientific civilization of the twentieth century, and science partly agrees with them. In reality, science is neither all paradox nor all animism, for it progresses in limited stages that are not foreseeable at too great a distance.

There exists in all the arts what we may call rationalism in the etymological sense: the search for proportion. The *artist* has always called upon it out of *necessity*. The rules of construction have varied widely over the centuries, but there have always been rules in every epoch because of the necessity of making oneself understood. Those who believe the first statement above are the first to refuse to apply the qualification *artistic* to a product which they do not *understand* at all.

Thus the musical scale is a convention which circumscribes the area of potentiality and permits construction within those limits in its own particular symmetry. The rules of Christian hymnography, of harmony, and of counterpoint in the various ages have allowed artists to construct and to make themselves understood by those who adopted the same constraints—through traditions, through collective taste or imitation, or through sympathetic resonance. The rules of serialism, for instance, those that banned the traditional octave doublings of tonality, imposed constraints which were partly new but none the less real.

Now everything that is rule or repeated constraint is part of the mental machine. A little “imaginary machine,” Philippot would have said—a choice, a set of decisions. A musical work can be analyzed as a multitude of mental machines. A melodic theme in a symphony is a mold, a mental machine, in the

same way as its structure is. These mental machines are something very restrictive and deterministic, and sometimes very vague and indecisive. In the last few years we have seen that this idea of mechanism is really a very general one. It flows through every area of human knowledge and action, from strict logic to artistic manifestations.

Just as the wheel was once one of the greatest products of human intelligence, a mechanism which allowed one to travel farther and faster with more luggage, so is the computer, which today allows the transformation of man's ideas. Computers resolve logical problems by heuristic methods. But computers are not really responsible for the introduction of mathematics into music; rather it is mathematics that makes use of the computer in composition. Yet if people's minds are in general ready to recognize the usefulness of geometry in the plastic arts (architecture, painting, etc.), they have only one more stream to cross to be able to conceive of using more abstract, non-visual mathematics and machines as aids to musical composition, which is more abstract than the plastic arts.

To summarize:

1. The creative thought of man gives birth to mental mechanisms, which, in the last analysis, are merely sets of constraints and choices. This process takes place in all realms of thought, including the arts.
2. Some of these mechanisms can be expressed in mathematical terms.
3. Some of them are physically realizable: the wheel, motors, bombs, digital computers, analogue computers, etc.
4. Certain mental mechanisms may correspond to certain mechanisms of nature.
5. Certain mechanizable aspects of artistic creation may be simulated by certain physical mechanisms or machines which exist or may be created.
6. It happens that computers can be useful in certain ways.

Here then is the theoretical point of departure for a utilization of electronic computers in musical composition.

We may further establish that the role of the living composer seems to have evolved, on the one hand, to one of inventing schemes (previously forms) and exploring the limits of these schemes, and on the other, to effecting the scientific synthesis of the new methods of construction and of sound emission. In a short while these methods must comprise all the ancient and modern means of musical instrument making, whether acoustic or electronic, with the help, for example, of digital-to-analogue converters; these have already been used in communication studies by N. Guttman, J. R. Pierce, and M. V. Mathews of Bell Telephone Laboratories in New Jersey. Now these explorations necessitate impressive mathematical, logical, physical, and psychological impedimenta, especially computers that accelerate the mental processes necessary for clearing the way for new fields by providing immediate experimental verifications at all stages of musical construction.

Music, by its very abstract nature, is the first of the arts to have attempted the conciliation of artistic creation with scientific thought. Its industrialization is inevitable and irreversible. Have we not already seen attempts to industrialize serial and popular music by the Parisian team of P. Barbaud, P. Blanchard, and Jeanine Charbonnier, as well as by the musicological research of Hiller and Isaacson at the University of Illinois?

In the preceding chapters we demonstrated some new areas of musical creation: Poisson, Markov processes, musical games, the thesis of the minimum of constraints, etc. They are all based on mathematics and especially on the theory of probability. They therefore lend themselves to being treated and

explored by computers. The simplest and most meaningful scheme is one of minimum constraints in composition, as exemplified by *Achorripsis*.

Thanks to my friend Georges Boudouris of the C.N.R.S. I made the acquaintance of Jacques Barraud, Engineer of the Ecole des Mines, then director of the Ensemble Electroniques de Gestion de la Société des Petroles Shell-Berre, and François Génuys, agrégé in mathematics, and head of the Etudes Scientifiques Nouvelles at IBM-France. All three are scientists, yet they consented to attempt an experiment which seemed at first far-fetched—that of a marriage of music with one of the most powerful machines in the world.

In most human relations it is rarely pure logical persuasion which is important; usually the paramount consideration is material interest. Now in this case it was not logic, much less self-interest, that arranged the betrothal, but purely experiment for experiment's sake, or game for game's sake, that induced collaboration. Stochastically speaking, my venture should have encountered failure. Yet the doors were opened, and at the end of a year and a half of contacts and hard work “the most unusual event witnessed by the firm or by this musical season [in Paris]” took place on 24 May 1962 at the headquarters of IBM-France. It was a live concert presenting a work of stochastic instrumental music entitled *ST/10-1, 080262*, which had been calculated on the IBM-7090. It was brilliantly performed by the conductor C. Simonovic and his Ensemble de Musique Contemporaine de Paris. By its passage through the machine, this work made tangible a stochastic method of composition, that of the minimum of constraints and rules.

Position of the Problem

The first working phase was the drawing up of the flow chart, i.e., writing down clearly and in order the stages of the operations of the scheme of *Achorripsis*,¹ and adapting it to the machine structure. In the first chapter we set out the entire synthetic method of this minimal structure. Since the machine is an iterative apparatus and performs these iterations with extraordinary speed, the thesis had to be broken down into a sequential series of operations reiterated in loops. An excerpt from the first flow chart is shown in Fig. V-1.

The statement of the thesis of *Achorripsis* receives its first machine-oriented interpretation in the following manner:

1. *The work consists of a succession of sequences or movements each a_i seconds long.* Their durations are totally independent (asymmetric) but have a fixed mean duration, which is introduced in the form of a parameter. These durations and their stochastic succession are given by the formula

$$P_{a_i} = ce^{-ca_i} da_i \quad (\text{See Appendix I.})$$

2. *Definition of the mean density of the sounds during a_i .* During a sequence sounds are emitted from several sonic sources. If the total number of these sounds or points during a sequence is N_{a_i} , the mean density of this point-cluster is N_{a_i}/a_i sounds/sec. In general, for a given instrumental ensemble this density has limits that depend on the number of instrumentalists, the nature of their instruments, and the technical difficulties of performance. For a large orchestra the upper limit is of the order of 150 sounds/sec. The lower limit (V3) is arbitrary and positive. We choose (V3) = 0.11 sounds/sec. Previous experiments led us to adopt a logarithmic progression for the density sensation with a number between 2 and 3 as its base. We adopted $e = 2.71827$. Thus the densities are included between $(V3)e^0$ and $(V3)e^R$ sounds/sec., which we can draw on a

¹See *Gravesaner Blätter*, nos. 11/12 (Mainz: Ars Viva Verlag, 1957).

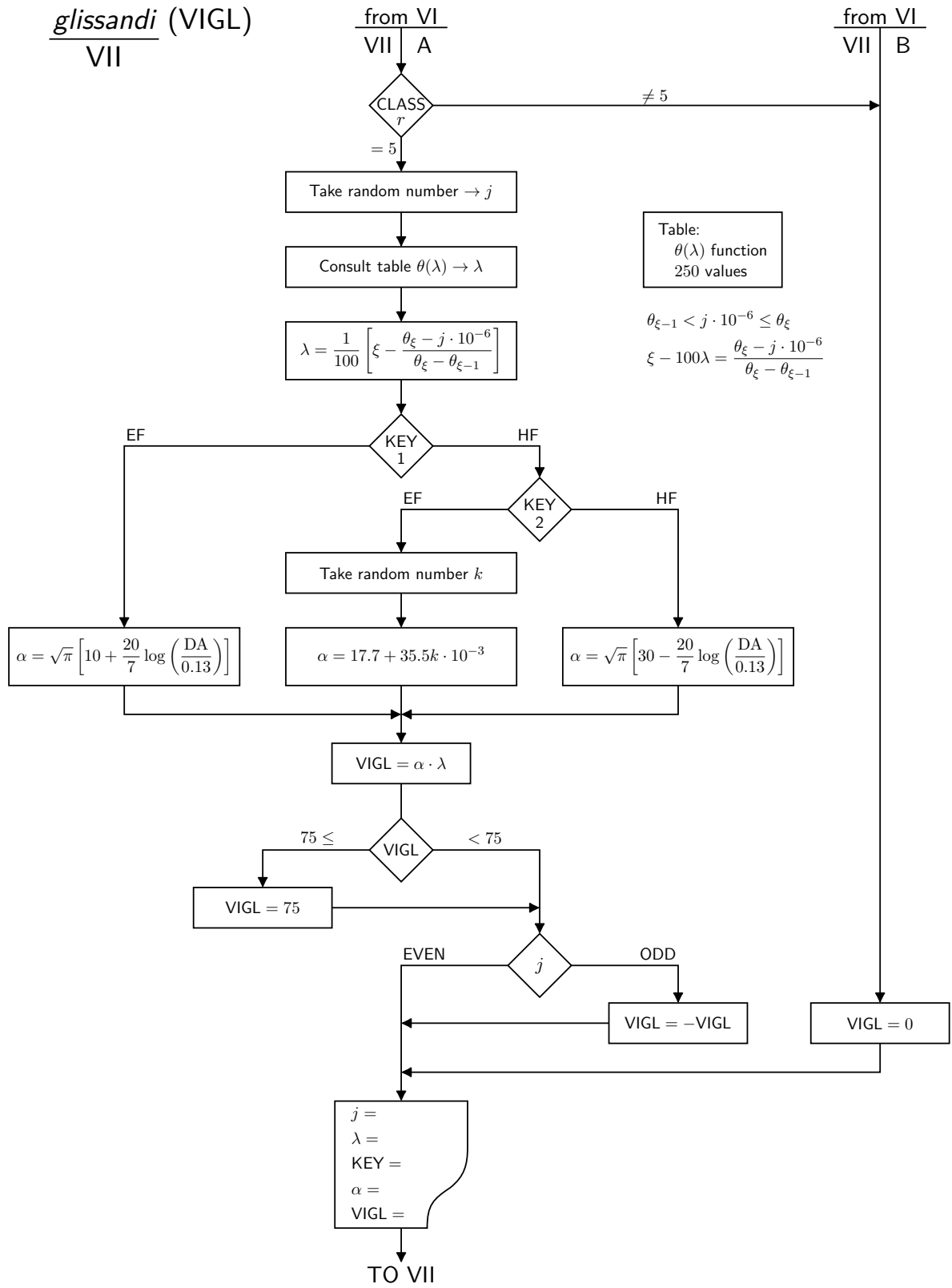


Figure 1: Excerpt from the First Flow Chart of Achorripsis

line graduated logarithmically (base e).² As our purpose is total independence, we attribute to each of the sequences a_i calculated in 1. a density represented by a point drawn at random from the portion of the line mentioned above. However a certain concern for continuity leads us to temper the independence of the densities among sequences a_i ; to this end we introduce a certain “memory” from sequence to sequence in the following manner:

Let a_{i-1} be a sequence of duration a_{i-1} , $(DA)_{i-1}$ its density, and a_i the next sequence with duration a_i and density $(DA)_i$. Density $(DA)_i$ will be given by the formula:

$$(DA)_i = (DA)_{i-1}e^{\pm x},$$

in which x is a segment of line drawn at random from a line segment s of length equal to $(R-0)$. The probability of x is given by

$$P_x = \frac{2}{s} \left(1 - \frac{x}{s}\right) dx \quad (\text{See Appendix I.})$$

and finally,

$$N_{a_i} = (DA)_i a_i.$$

3. *Composition Q of the orchestra during sequence a_i .* First the instruments are divided into r classes of timbres, e.g., flutes and clarinets, oboes and bassoons, brasses, bowed strings, pizzicati, col legno strokes, glissandi, wood, skin, and metal percussion instruments, etc. (See the table for *Atrées*.) The composition of the orchestra is stochastically conceived, i.e., the distribution of the classes is not deterministic. Thus during a sequence of duration a_i it may happen that we have 80% pizzicati, 10% percussion, 7% keyboard, and 3% flute class. Under actual conditions the determining factor which would condition the composition of the orchestra is density. We therefore connect the orchestral composition with density by means of a special diagram. An example from *ST/10-1, 080262* is shown in Fig. V-2.

Fig. V-2 is expressed by the formula

$$Q_r = (n-x)(e_{n,r} - e_{n+1,r}) + e_{n,r}$$

in which r = the number of the class, $x = \log_e [(DA)_i / (V3)]$, $n = 0, 1, 2, \dots, R$, such that $n \leq x \leq n+1$, and $e_{n,r}$ and $e_{n+1,r}$ are the probabilities of class r as a function of n . It goes without saying that the composition of this table is a precise task of great complexity and delicacy. Once these preliminaries have been completed, we can define, one after the other, the N_{a_i} sounds of sequence a_i .

4. *Definition of the moment of occurrence of the sound N within the sequence a_i .* The mean density of the points or sounds to be distributed within a_i is $k = N_{a_i} / a_i$. The formula which gives the intervals separating the sound attacks is

$$P_t = ke^{-kt} dt. \quad (\text{See Appendix I.})$$

5. *Attribution to the above sound of an instrument belonging to orchestra Q , which has already been calculated.* First class r is drawn at random with probability q_r from the orchestra ensemble calculated in 3. (Consider an urn with balls of r colors in various proportions.) Then from within class r the number of the instrument is drawn according to the probability p_n given by an arbitrary table (urn with balls of n colors). Here also the distribution of instruments within a class is delicate and complex.

² $(V3)e^R$ must be equal to the upper limit, e.g., to 150 sounds/sec. in the case of a large orchestra.

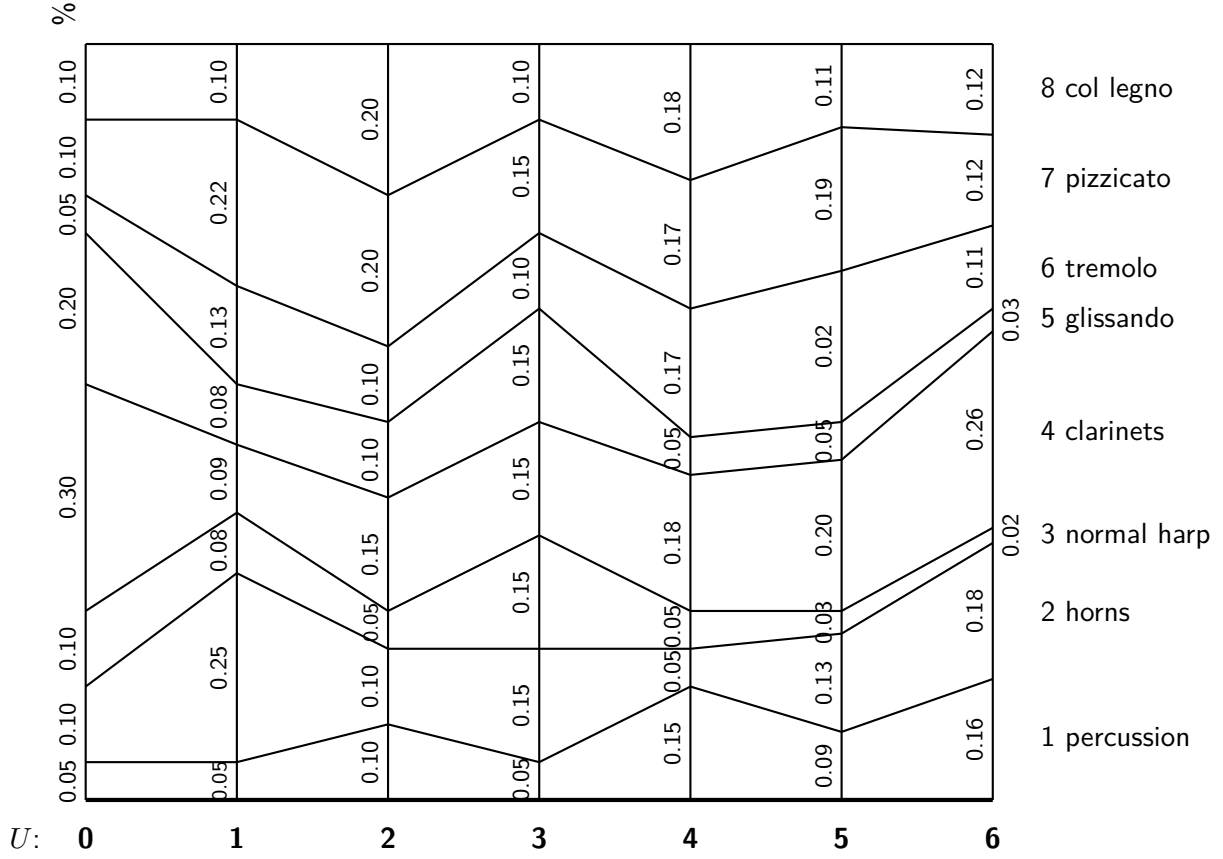


Figure 2: *ST/10-1, 080262*, Composition of the Orchestra

6. *Attribution of a pitch as a function of the instrument.* Taking as the zero point the lowest Bb of the piano, we establish a chromatic scale in semitones of about 85 degrees. The range s of each instrument is thus expressed by a natural number (distance). But the pitch h_u of a sound is expressed by a decimal number of which the whole number part is related to a note of the chromatic scale within the instrument's range.

Just as for the density in 2., we accept a certain memory of or dependence on the preceding pitch played by the same instrument, so that we have

$$h_u = h_{u-1} \pm z,$$

where z is given by the probability formula

$$P_z = \frac{2}{s} \left(1 - \frac{z}{s}\right) dz. \quad (\text{See Appendix I.})$$

P_z is the probability of the interval z taken at random from the range s , and s is expressed as the difference between the highest and lowest pitches that can be played on the instrument.

7. *Attribution of a glissando speed if class r is characterized as a glissando.* The homogeneity hypotheses in Chap. I led us to the formula

$$f(v) = \frac{2}{a\sqrt{\pi}} e^{-v^2/a^2},$$

and by the transformation $v/a = u$ to its homologue:

$$T(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-u^2} du,$$

for which there are tables. $f(v)$ is the probability of occurrence of the speed v (which is expressed in semitones/sec.); it has a parameter a , which is proportional to the standard deviation s ($a = s\sqrt{2}$). a is defined as a function of the logarithm of the density of sequence a_i by: an inversely proportional function

$$a = \sqrt{\pi} \left(30 - \frac{20}{R} \log [(DA)_i / (V3)] \right),$$

or a directly proportional function

$$a = \sqrt{\pi} \left(10 + \frac{20}{R} \log [(DA)_i / (V3)] \right),$$

or a function independent of density

$$a = 17.7 + 35k,$$

where k is a random number between 0 and 1.

The constants of the preceding formulae derive from the limits of the speeds that string glissandi may take.

Thus for $(DA)_i = 145$ sounds/sec.

$$a = 53.2 \text{ semitones/sec.}$$

$$2s = 75 \text{ semitones/sec.}$$

and for $(DA)_i = 0.13$ sounds/sec.

$$a = 17.7 \text{ semitones/sec.}$$

$$2s = 25 \text{ semitones/sec.}$$

8. *Attribution of a duration x to the sounds emitted.* To simplify we establish a mean duration for each instrument, which is independent of tessitura and nuance. Consequently we reserve the right to modify it when transcribing into traditional notation. The following is the list of constraints that we take into account for the establishment of duration x :

G , the maximum length of respiration or desired duration

$(DA)_i$, the density of the sequence

q_r , the probability of class r

p_n , the probability of the instrument n

Then if we define z as a parameter of a sound's duration, z could be inversely proportional to the probability of the occurrence of the instrument, so that

$$z = \frac{1}{(DA)_i p_n q_r}.$$

z will be at its maximum when $(DA)_i p_n q_r$ is at its minimum, and in this case we could choose $z_{\max} = G$.

Instead of letting $z_{\max} = G$, we shall establish a logarithmic law so as to freeze the growth of z . This law applies for any given value of z .

$$z' = G \ln z / \ln z_{\max}$$

Since we admit a total independence, the distribution of the durations x will be Gaussian:

$$f(x) = \frac{1}{s\sqrt{2\pi}} e^{-(x-m)^2/(2s^2)},$$

where m is the arithmetic mean of the durations, s the standard deviation, and

$$m - 4.25s = 0$$

$$m + 4.25s = z'$$

the linear system which furnishes us with the constants m and s . By assuming $u = (x - m)/s\sqrt{2}$ we find the function $T(u)$, for which we consult the tables.

Finally, the duration x of the sound will be given by the relation

$$x = \pm us\sqrt{2} + m.$$

We do not take into account incompatibilities between instruments, for this would needlessly burden the machine's program and calculation.

9. *Attribution of dynamic forms to the sounds emitted.* We define four zones of mean intensities: *ppp*, *p*, *f*, *ff*. Taken three at a time they yield $4^3 = 64$ permutations, of which 44 are different (an urn with 44 colors); for example, ppp <f> p. <!-- Table of the 44 Intensity Forms Derived from 4 Mean Intensity Values, ppp, p, f, ff -->
10. *The same operations are begun again for each sound of the cluster N_{a_i} .*
11. *Recalculations of the same sort are made for the other sequences.*

An extract from the sequential statement was reproduced in Fig. V-1. Now we must proceed to the transcription into Fortran IV, a language "understood" by the machine (see Fig. V-3). <!-- Fig. V-3. Stochastic Music Rewritten in Fortran IV --> <!-- Fig. V-4. Provisional Results of One Phase of the Analysis --> <!-- Fig. V-5. Bars 1-5 of ST/10-1,080262 -->

It is not our purpose to describe the transformation of the flow chart into Fortran. However, it would be interesting to show an example of the adaptation of a mathematical expression to machine methods.

Let us consider the elementary law of probability (density function)

$$f(x) dx = ce^{-cx} dx.$$

How shall we proceed in order for the computer to give us lengths x with the probability $f(x) dx$? The machine can only draw random numbers y_0 with equiprobability between 0 and 1. We shall "modulate" this probability: Assume some length x_0 ; then we have

$$\text{prob}(0 \leq x \leq x_0) = \int_0^{x_0} f(x) dx = 1 - e^{-cx_0} = F(x_0),$$

where $F(x_0)$ is the distribution function of x . But

$$F(x_0) = \text{prob}(0 \leq y \leq y_0) = y_0$$

then

$$1 - e^{-cx_0} = y_0$$

and

$$x_0 = -\frac{\ln(1 - y_0)}{c}$$

for all $x_0 \geq 0$.

Once the program is transcribed into language that the machine's internal organization can assimilate, a process that can take several months, we can proceed to punching the cards and setting up certain tests. Short sections are run on the machine to detect errors of logic and orthography and to determine the values of the entry parameters, which are introduced in the form of variables. This is a very important phase, for it permits us to explore all parts of the program and determine the modalities of its operation. The final phase is the decoding of the results into traditional notation, unless an automatic transcriber is available.

Conclusions

A large number of compositions of the same kind as *ST/10-1, 080262* is possible for a large number of orchestral combinations. Other works have already been written: *ST/48-1, 240162*, for large orchestra, commissioned by RTF (France III); *Atrées* for ten soloists; and *Morisma-Amorisima*, for four soloists.

Although this program gives a satisfactory solution to the minimal structure, it is, however, necessary to jump to the stage of pure composition by coupling a digital-to-analogue converter to the computer. The numerical calculations would then be changed into sound, whose internal organization had been conceived beforehand. At this point one could bring to fruition and generalize the concepts described in the preceding chapters.

The following are several of the advantages of using electronic computers in musical composition:

1. The long laborious calculation made by hand is reduced to nothing. The speed of a machine such as the IBM-7090 is tremendous—of the order of 500,000 elementary operations/sec.
2. Freed from tedious calculations the composer is able to devote himself to the general problems that the new musical form poses and to explore the nooks and crannies of this form while modifying the values of the input data. For example, he may test all instrumental combinations from soloists to chamber orchestras, to large orchestras. With the aid of electronic computers the composer becomes a sort of pilot: he presses the buttons, introduces coordinates, and supervises the controls of a cosmic vessel sailing in the space of sound, across sonic constellations and galaxies that he could formerly glimpse only as a distant dream. Now he can explore them at his ease, seated in an armchair.
3. The program, i.e., the list of sequential operations that constitute the new musical form, is an objective manifestation of this form. The program may consequently be dispatched to any point on the earth that possesses computers of the appropriate type, and may be exploited by any composer pilot.
4. Because of certain uncertainties introduced in the program, the composer-pilot can instill his own personality in the sonic result he obtains.

C	PROGRAM FREE STOCHASTIC MUSIC (FORTRAN IV)	XEN	6
C		XEN	7
C	GLOSSARY OF THE PRINCIPAL ABBREVIATIONS	XEN	8

C			
C	A - DURATION OF EACH SEQUENCE IN SECONDS	XEN	9
C	A10,A20,A17,A35,A30 - NUMBERS FOR GLISSANDO CALCULATION	XEN	10
C	ALEA - PARAMETER USED TO ALTER THE RESULT OF A SECOND RUN WITH THE	XEN	11
C	SAME INPUT DATA	XEN	12
C	ALFA(3) - THREE EXPRESSIONS ENTERING INTO THE THREE SPEED VALUES	XEN	13
C	OF THE SLIDING TONES (GLISSANDI)	XEN	14
C	ALIM - MAXIMUM LIMIT OF SEQUENCE DURATION A	XEN	15
C	(AMAX(I),I=1,KTR) TABLE OF AN EXPRESSION ENTERING INTO THE	XEN	16
C	CALCULATION OF THE NOTE LENGTH IN PART 8	XEN	17
C	BF - DYNAMIC FORM NUMBER, THE LIST IS ESTABLISHED INDEPENDENTLY	XEN	18
C	OF THIS PROGRAM AND IS SUBJECT TO MODIFICATION	XEN	19
C	DELTA - THE RECIPROCAL OF THE MEAN DENSITY OF SOUND EVENTS DURING	XEN	20
C	A SEQUENCE OF DURATION A	XEN	21
C	(E(I,J),I=1,KTR,J=1,KTE) - PROBABILITIES OF THE KTR TIMBRE CLASSES	XEN	22
C	INTRODUCED AS INPUT DATA, DEPENDING ON THE CLASS NUMBER I=KR	XEN	23
C	AND ON THE POWER J=U OBTAINED FROM $V3*EXP(U)=DA$	XEN	24
C	EPSI - EPSILON FOR ACCURACY IN CALCULATING PN AND E(I,J), WHICH	XEN	25
C	IT IS ADVISABLE TO RETAIN.	XEN	26
C	(GN(I,J),I=1,KTR,J=1,KTS) - TABLE OF THE GIVEN LENGTH OF BREATH	XEN	27
C	FOR EACH INSTRUMENT, DEPENDING ON CLASS I AND INSTRUMENT J	XEN	28
C	GTNA - GREATEST NUMBER OF NOTES IN THE SEQUENCE OF DURATION A	XEN	29
C	GTNS - GREATEST NUMBER OF NOTES IN KW LOOPS	XEN	30
C	(HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),I=1,KTR,J=1,KTS)	XEN	31
C	TABLE OF INSTRUMENT COMPASS LIMITS, DEPENDING ON TIMBRE CLASS I	XEN	32
C	AND INSTRUMENT J. TEST INSTRUCTION 480 IN PART 6 DETERMINES	XEN	33
C	WHETHER THE HA OR THE HB TABLE IS FOLLOWED. THE NUMBER 7 IS	XEN	34
C	ARBITRARY.	XEN	35
C	JW - ORDINAL NUMBER OF THE SEQUENCE COMPUTED.	XEN	36
C	KNL - NUMBER OF LINES PER PAGE OF THE PRINTED RESULT.KNL=50	XEN	37
C	KR1 - NUMBER IN THE CLASS KR=1 USED FOR PERCUSSION OR INSTRUMENTS	XEN	38
C	WITHOUT A DEFINITE PITCH.	XEN	39
C	KTE - POWER OF THE EXPONENTIAL COEFFICIENT E SUCH THAT	XEN	40
C	$DA(MAX)=V3*(E**(KTE-1))$	XEN	41
C	KTR - NUMBER OF TIMBRE CLASSES	XEN	42
C	KW - MAXIMUM NUMBER OF JW	XEN	43
C	KTEST1,TAV1,ETC. - EXPRESSIONS USEFUL IN CALCULATING HOW LONG THE	XEN	44
C	VARIOUS PARTS OF THE PROGRAM WILL RUN.	XEN	45
C	KT1 - ZERO IF THE PROGRAM IS BEING RUN, NONZERO DURING DEBUGGING	XEN	46
C	KT2 - NUMBER OF LOOPS, EQUAL TO 15 BY ARBITRARY DEFINITION.	XEN	47
C	(MODI(IX8),IX8=7,1) AUXILIARY FUNCTION TO INTERPOLATE VALUES IN	XEN	48
C	THE TETA(256) TABLE (SEE PART 7)	XEN	49
C	NA - NUMBER OF SOUNDS CALCULATED FOR THE SEQUENCE A (NA=DA*A)	XEN	50
C	(NT(I),I=1,KTR) NUMBER OF INSTRUMENTS ALLOCATED TO EACH OF THE	XEN	51
C	KTR TIMBRE CLASSES.	XEN	52
C	(PN(I,J),I=1,KTR,J=1,KTS),(KTS=NT(I),I=1,KTR) TABLE OF PROBABILITY	XEN	53
C	OF EACH INSTRUMENT OF THE CLASS I.	XEN	54
C	(Q(I),I=1,KTR) PROBABILITIES OF THE KTR TIMBRE CLASSES, CONSIDERED	XEN	55
C	AS LINEAR FUNCTIONS OF THE DENSITY DA.	XEN	56
C	(S(I),I=1,KTR) SUM OF THE SUCCESSIVE Q(I) PROBABILITIES, USED TO	XEN	57
C	CHOOSE THE CLASS KR BY COMPARING IT TO A RANDOM NUMBER X1 (SEE	XEN	58
C	PART 3, LOOP 380 AND PART 5, LOOP 430).	XEN	59
C	SINA - SUM OF THE COMPUTED NOTES IN THE NEW CLOUDS NA, ALWAYS LESS	XEN	60
C	THAN GTNS (SEE TEST IN PART 10).	XEN	61
C	SQPI - SQUARE ROOT OF PI (3.14159...)	XEN	62
C	TA - SOUND ATTACK TIME ABCISSA.	XEN	63
C	TETA(256) - TABLE OF THE 256 VALUES OF THE INTEGRAL OF THE NORMAL	XEN	64
C	DISTRIBUTION CURVE WHICH IS USEFUL IN CALCULATING GLISSANDO SPEED	XEN	65
C	AND SOUND EVENT DURATION.	XEN	66

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C      VIGL - GLISSANDO SPEED (VITESSE GLISSANDO), WHICH CAN VARY AS, BE XEN 67
C      INDEPENDENT OF, OR VARY INVERSELY AS THE DENSITY OF THE SEQUENCE, XEN 68
C      THE ACTUAL MODE OF VARIATION EMPLOYED REMAINING THE SAME FOR THE XEN 69
C      ENTIRE SEQUENCE (SEE PART 7). XEN 70
C      VITLIM - MAXIMUM LIMITING GLISSANDO SPEED (IN SEMITONES/SEC), XEN 71
C      SUBJECT TO MODIFICATION. XEN 72
C      V3 - MINIMUM CLOUD DENSITY DA XEN 73
C      (Z1(I),Z2(I),I=1,8) TABLE COMPLEMENTARY TO THE TETA TABLE. XEN 74
C XEN 75
C XEN 76
C      READ CONSTANTS AND TABLES XEN 78
C XEN 77
      DIMENSION Q(12),S(12),E(12,12),PN(12,50),SPN(12,50),NT(12), XEN 79
1HAMIN(12,50),HAMAX(12,50),HBMIN(12,50),HBMAX(12,50),GN(12,50),H(12,XEN 80
2,50),TETA(256),VIGL(3),MODI(7),Z1(8),Z2(8),ALFA(3),AMAX(12) XEN 81
C XEN 82
C XEN 83
C XEN 84
      I=1 XEN 85
      DO 10 IX=1,7 XEN 86
      IX8=8-IX XEN 87
      MODI(IX8)=I XEN 88
10      I=I+1 XEN 89
C XEN 90
      READ 20,(TETA(I),I=1,256) XEN 91
20      FORMAT(12F6.6) XEN 92
      READ 30,(Z1(I),Z2(I),I=1,8) XEN 93
30      FORMAT(6(F3.2,F9.8)/F3.2,F9.8,E6.2,F9.8) XEN 94
      PRINT 40,TETA,Z1,Z2 XEN 95
40      FORMAT('1THE TETA TABLE= ',/,21(12F10.6,/),4F10.6,/////, XEN 96
1' THE Z1 TABLE= ',/,7F6.2,E12.3,///, ' THE Z2 TABLE= ',/,8F14.8,/XEN 97
21H1) XEN 98
      READ 50,DELTA,V3,A10,A20,A17,A30,A35,BF,SQPI,EPSI,VITLIM,ALEA, AXEN 99
1LIM XEN 100
50      FORMAT(F3.0,F3.3,5F3.1,F2.0,F8.7,F8.8,F4.2,F8.8,F5.2) XEN 114
      READ 60,KT1,KT2,KW,KNL,KTR,KTE,KR1,GTNA,GTNS,(NT(I),I=1,KTR) XEN 115
60      FORMAT(5I3,2I2,2F6.0,12I2) XEN 126
      PRINT 70,DELTA,V3,A10,A20,A17,A30,A35,BF,SQPI,EPSI,VITLIM,ALEA, AXEN 127
1LIM,KT1,KT2,KW,KNL,KTR,KTE,KR1,GTNA,GTNS,(I,NT(I),I=1,KTR) XEN 128
70      FORMAT('1DELTA= ',F4.0,/,' V3= ',F6.3,/,' A10= ',F4.1,/,' XEN 129
1' A20= ',F4.1,/,' A17= ',F4.1,/,' A30= ',F4.1,/,' A35= ',F4.1,XEN 130
2/,' BF= ',F3.0,/,' SQPI= ',F11.8,/,' EPSI= ',F12.8,/,' VITLIM= ' XEN 131
3,F5.2,/,' ALEA= ',F12.8,/,' ALIM= ',F6.2,/,' KT1= ',I3,/,' XEN 132
4' KT2= ',I3,/,' KW= ',I3,/,' KNL= ',I3,/,' KTR= ',I3,/,' XEN 133
5' KTE= ',I2,/,' KR1= ',I2,/,' GTNA= ',F7.0,/,' GTNS= ',F7.0, XEN 134
6/,' 12( ' IN CLASS ',I2,' , THERE ARE ',I2,' INSTRUMENTS. ',/)) XEN 135
      READ 80,KTEST3,KTEST1,KTEST2 XEN 136
80      FORMAT(5I3) XEN 141
      PRINT 90,KTEST3,KTEST1,KTEST2 XEN 142
90      FORMAT(' KTEST3= ',I3,/,' KTEST1= ',I3,/,' KTEST2= ',I3) XEN 143
C XEN 144
      IF(KTEST3.NE.0) PRINT 830 XEN 145
      R=KTE-1 XEN 146
      A10=A10*SQPI XEN 147
      A20=A20*SQPI/R XEN 148
      A30=A30*SQPI XEN 149
C      IF ALEA IS NON-ZERO, THE RANDOM NUMBER IS GENERATED FROM THE TIME XEN 150
C      WHEN THE FOLLOWING INSTRUCTION IS EXECUTED. IF ALEA IS NON-ZERO XEN 151
C      EACH RUN OF THIS PROGRAM WILL PRODUCE DIFFERENT OUTPUT DATA. XEN 152

```

IF(ALEA.NE.0.0) CALL RANFSET(TIMEF(1))	XEN	153
PRINT 830	XEN	154
DO 130 I=1,KTR	XEN	155
Y=0.0	XEN	156
KTS=NT(I)	XEN	157
READ 100,(HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),GN(I,J),	XEN	158
1PN(I,J),J=1,KTS)	XEN	159
100 FORMAT(5(5F2.0,F3.3))	XEN	160
PRINT 110,I,(J,HAMIN(I,J),HAMAX(I,J),HBMIN(I,J),HBMAX(I,J),GN(I,J)	XEN	161
1,PN(I,J),J=1,KTS)	XEN	162
110 FORMAT(////,'INCLASSNUMBER',I2,/,('FORINSTRUMENTNO.',I2,	XEN	163
1'HAMIN=',F3.0,'HAMAX=',F3.0,'HBMIN=',F3.0,'HBMAX=',	XEN	164
2 F3.0,'GN=',F3.0,'ANDPN=',F6.3))	XEN	165
DO 120 J=1,KTS	XEN	166
Y=Y+PN(I,J)	XEN	167
120 SPN(I,J)=Y	XEN	168
130 IF(ABSF(Y-1.0).GE.EPSI) CALL EXIT	XEN	169
C	XEN	170
DO 150 I=1,KTR	XEN	171
READ 140,(E(I,J),J=1,KTE)	XEN	172
140 FORMAT(12F2.2)	XEN	173
150 PRINT 160,I,(J,E(I,J),J=1,KTE)	XEN	174
160 FORMAT(////////,'CLASSNUMBER',I2,/,('INDENSITYLEVEL',I2,	XEN	175
1'HASAPROBABILITYOF',F6.2))	XEN	176
DO 180 J=1,KTE	XEN	177
Y=0.0	XEN	178
DO 170 I=1,KTR	XEN	179
170 Y=Y+E(I,J)	XEN	180
180 IF(ABSF(Y-1.0).GE.EPSI) CALL EXIT	XEN	181
DO 200 I=1,KTR	XEN	182
AMAX(I)=1.0/E(I,1)	XEN	183
DO 200 J=2,KTE	XEN	184
AJ=J-1	XEN	185
AX=1.0/(E(I,J)*EXPF(AJ))	XEN	186
IF (KT1.NE.0) PRINT 190,AX	XEN	187
190 FORMAT(1H ,9E12.8)	XEN	188
200 IF(AX.GT.AMAX(I)) AMAX(I)=AX	XEN	189
IF(KT1.NE.0) WRITE(2,210) AMAX	XEN	190
210 FORMAT(6E15.8)	XEN	191
C	XEN	192
JW=1	XEN	193
SINA=0.0	XEN	194
IF(KTEST1.NE.0) TAV1=TIMEF(1)	XEN	195
220 NLINE=50	XEN	196
C	XEN	197
C PARTS 1 AND 2, DEFINE SEQUENCE A SECONDS AND CLOUD NA DURING A	XEN	198
C	XEN	199
KNA=0	XEN	200
K1=0	XEN	201
230 X1=RANF(-1)	XEN	202
A=-DELTA * LOGF(X1)	XEN	203
IF(A.LE.ALIM) GO TO 250	XEN	204
IF(K1.GE.KT2) GO TO 240	XEN	205
K1=K1+1	XEN	206
GO TO 230	XEN	207
240 A=ALIM/2.0	XEN	208
X1=0.0	XEN	209
250 K2=0	XEN	210
260 X2=RANF(-1)	XEN	211

	IF(JW.GT.1) GO TO 280	XEN	212
270	UX=R*X2	XEN	213
	GO TO 310	XEN	214
280	IF(RANF(-1).GE.0.5) GO TO 290	XEN	215
	UX=UPR + R * (1.0-SQRTF(X2))	XEN	216
	GO TO 300	XEN	217
290	UX=UPR - R * (1.0-SQRTF(X2))	XEN	218
300	IF((UX.GE.0.0).AND.(UX.LE.R)) GO TO 310	XEN	219
	IF(K2.GE.KT2) GO TO 270	XEN	220
	K2=K2+1	XEN	221
	GO TO 260	XEN	222
310	U=UX	XEN	223
	DA=V3*EXPF(U)	XEN	224
	NA=XINTF(A * DA + 0.5) + 1	XEN	225
	IF (GTNA.GT.FLOATF(NA)) GO TO 330	XEN	226
	IF (KNA.GE.KT2) GO TO 320	XEN	227
	KNA=KNA+1	XEN	228
	GO TO 230	XEN	229
320	A=DELTA	XEN	230
	GO TO 260	XEN	231
330	UPR=U	XEN	232
	IF (KT1.EQ.0) GO TO 360	XEN	233
	PRINT 340,JW,KNA,K1,K2,X1,X2,A,DA,NA	XEN	234
340	FORMAT(1H1,4I8,3X,4E18.8,3X,I8)	XEN	235
	NA=KT1	XEN	236
	IF (KTEST3.NE.0) PRINT 350,JW,NA,A	XEN	237
350	FORMAT(1H0,2I9,F10.2)	XEN	238
C		XEN	239
C	PART 3. DEFINE CONSTITUTION OF ORCHESTRA DURING SEQUENCE A	XEN	240
C		XEN	241
360	SINA=SINA + FLOATF(NA)	XEN	242
	XLOGDA=U	XEN	243
	XALOG=A20 *XLOGDA	XEN	244
	M=XINTF(XLOGDA)	XEN	245
	IF ((M+2).GT.KTE) M=KTE-2	XEN	246
	SR=0.0	XEN	247
	M1=M+1	XEN	248
	M2=M+2	XEN	249
	DO 380 I=1,KTR	XEN	250
	ALFX=E(I,M1)	XEN	251
	BETA=E(I,M2)	XEN	252
	XM=M	XEN	253
	QR=(XLOGDA-XM) * (BETA-ALFX) + ALFX	XEN	254
	IF (KT1.NE.0) PRINT 370,XM,ALFX,BETA	XEN	255
370	FORMAT(1H ,3F20.8)	XEN	256
	Q(I)=QR	XEN	257
	SR=SR+QR	XEN	258
380	S(I)=SR	XEN	259
	IF (KT1.NE.0) PRINT 390,(Q(I),I=1,KTR),(S(I),I=1,KTR)	XEN	260
390	FORMAT(1H ,12F9.4)	XEN	261
C		XEN	262
C	PART 4,DEFINE INSTANT TA OF EACH POINT IN SEQUENCE A	XEN	263
C		XEN	264
	IF (KTEST2.NE.0) TAV2=TIMEF(1)	XEN	265
	N=1	XEN	266
	T=0.0	XEN	267
	TA=0.0	XEN	268
	GO TO 410	XEN	269
400	N=N+1	XEN	270

X=RANF(-1)	XEN	271
T=-LOGF(X)/DA	XEN	272
TA=TA+T	XEN	273
410 IF (KT1.NE.0) PRINT 420,N,X,T,TA	XEN	274
420 FORMAT(//,I8,3E20.8)	XEN	275
C	XEN	276
C PART 5,DEFINE CLASS AND INSTRUMENT NUMBER TO EACH POINT OF A	XEN	277
C	XEN	278
X1=RANF(-1)	XEN	279
DO 430 I=1,KTR	XEN	280
430 IF (X1.LE.S(I)) GO TO 440	XEN	281
I=KTR	XEN	282
440 KTS=NT(I)	XEN	283
KR=I	XEN	284
X2=RANF(-1)	XEN	285
DO 450 J=1,KTS	XEN	286
SPIEN=SPN(KR,J)	XEN	287
INSTRM=J	XEN	288
450 IF (X2.LE.SPIEN) GO TO 460	XEN	289
INSTRM=KTS	XEN	290
460 PIEN=PN(KR,INSTRM)	XEN	291
IF (KT1.NE.0) PRINT 470,X1,S(KR),KR,X2,SPIEN,INSTRM	XEN	292
470 FORMAT(1H ,2E20.8,I6,2E20.8,I6)	XEN	293
C	XEN	294
C PART 6,DEFINE PITCH HN FOR EACH POINT OF SEQUENCE A	XEN	295
C	XEN	296
IF (KR.GT.1) GO TO 480	XEN	297
IF (INSTRM.GE.KR1) GO TO 490	XEN	298
HX=0.0	XEN	299
GO TO 560	XEN	300
480 IF (KR.LT.7) GO TO 490	XEN	301
HSUP=HBMAX(KR,INSTRM)	XEN	302
HINF=HBMIN(KR,INSTRM)	XEN	303
GO TO 500	XEN	304
490 HSUP=HAMAX(KR,INSTRM)	XEN	305
HINF=HAMIN(KR,INSTRM)	XEN	306
500 HM=HSUP-HINF	XEN	307
HPR=H(KR,INSTRM)	XEN	308
K=0	XEN	309
IF (HPR.LE.0.0) GO TO 520	XEN	310
510 X=RANF(-1)	XEN	311
IF (N.GT.1) GO TO 530	XEN	312
520 HX=HINF+HM*X RANF(-1)	XEN	313
GO TO 560	XEN	314
530 IF(RANF(-1).GE.0.5) GO TO 540	XEN	315
HX=HPR+HM * (1.0-SQRTF(X))	XEN	316
GO TO 550	XEN	317
540 HX=HPR-HM * (1.0-SQRTF(X))	XEN	318
550 IF((HX.GE.HINF).AND.(HX.LE.HSUP)) GO TO 560	XEN	319
IF (K.GE.KT2) GO TO 520	XEN	320
K=K+1	XEN	321
GO TO 510	XEN	322
560 H(KR,INSTRM)=HX	XEN	323
IF (KT1.NE.0) PRINT 570,K,X,HX	XEN	324
570 FORMAT(1H ,I6,2E20.8)	XEN	325
C	XEN	326
C PART 7,DEFINE SPEED VIGL TO EACH POINT OF A	XEN	327
C	XEN	328
IF (KR.EQ.5) GO TO 580	XEN	329

VIGL(1)=0.0	XEN	330
VIGL(2)=0.0	XEN	331
VIGL(3)=0.0	XEN	332
X1=0.0	XEN	333
X2=0.0	XEN	334
XLAMBDA=0.0	XEN	335
GO TO 740	XEN	336
580 KX=1	XEN	337
590 X1=RANF(-1)	XEN	338
IF (X1-0.9997) 600,650,680	XEN	339
600 I=128	XEN	340
DO 630 IX=1,7	XEN	341
IF(TETA(I)-X1) 610,640,620	XEN	342
610 I=I+MODI(IX)	XEN	343
GO TO 630	XEN	344
620 I=I-MODI(IX)	XEN	345
630 CONTINUE	XEN	346
IF(TETA(I)-X1) 670,640,660	XEN	347
640 XLAMBDA=FLOATF(I-1)/100.0	XEN	348
GO TO (720,760), KX	XEN	349
650 XLAMBDA=2.55	XEN	350
GO TO (720,760), KX	XEN	351
660 I=I-1	XEN	352
670 TX1=TETA(I)	XEN	353
XLAMBDA=(FLOATF(I-1)+(X1-TX1)/(TETA(I+1)-TX1))/100.0	XEN	354
GO TO (720,760), KX	XEN	355
680 DO 690 I=2,7	XEN	356
TX1=Z2(I)	XEN	357
IF(X1-TX1) 700,710,690	XEN	358
690 CONTINUE	XEN	359
I=8	XEN	360
TX1=1.0	XEN	361
700 TX2=Z1(I)	XEN	362
XLAMBDA=TX2-((TX1-X1)/(TX1-Z2(I-1)))*(TX2-Z1(I-1))	XEN	363
GO TO (720,760), KX	XEN	364
710 XLAMBDA=Z1(I)	XEN	365
GO TO(720,760), KX	XEN	366
720 ALFA(1)=A10+XALOG	XEN	367
ALFA(3)=A30-XALOG	XEN	368
X2=RANF(-1)	XEN	369
ALFA(2)=A17+A35*X2	XEN	370
DO 730 I=1,3	XEN	371
VIGL(I)=INTF(ALFA(I)*XLAMBDA+0.5)	XEN	372
IF (VIGL(I).LT.0.0) VIGL(I)=-VIGL(I)	XEN	373
IF (VIGL(I).GT.VITLIM) VIGL(I)=VITLIM	XEN	374
730 IF (RANF(-1).LT.0.5) VIGL(I)=-VIGL(I)	XEN	375
740 IF(KT1.NE.0) PRINT 750,X1,X2,XLAMBDA,VIGL	XEN	376
750 FORMAT(1H ,6E19.8)	XEN	377
C	XEN	378
C PART 8,DEFINE DURATION FOR EACH POINT OF A	XEN	379
C	XEN	380
IF ((KR.EQ.7).OR.(KR.EQ.8)) GO TO 780	XEN	381
ZMAX=AMAX(KR)/(V3*PIEN)	XEN	382
G=GN(KR, INSTRM)	XEN	383
RO=G/LOGF(ZMAX)	XEN	384
QPNDA=1.0/(Q(KR)*PIEN*DA)	XEN	385
GE=ABSF(RO*LOGF(QPNDA))	XEN	386
XMU=GE/2.0	XEN	387
SIGMA=GE/4.0	XEN	388

KX=2	XEN	389
GO TO 590	XEN	390
760 TAU=SIGMA*XLAMBDA*1.4142	XEN	391
X2=RANF(-1)	XEN	392
IF (X2.GE.0.5) GO TO 770	XEN	393
XDUR=XMU+TAU	XEN	394
GO TO 790	XEN	395
770 XDUR=XMU-TAU	XEN	396
IF (XDUR.GE.0.0) GO TO 790	XEN	397
780 XDUR=0.0	XEN	398
790 IF(KT1.NE.0)PRINT 800,ZMAX,XMU,SIGMA,X1,XLAMBDA,X2,XDUR	XEN	399
800 FORMAT(1H ,5E15.8,E11.4,E15.8)	XEN	400
C	XEN	401
C PART 9,DEFINE INTENSITY FORM TO EACH POINT OF A	XEN	402
C	XEN	403
IFORM=XINTF(RANF(-1)*BF+0.5)	XEN	404
IF (KT1.EQ.0) GO TO 840	XEN	405
IF (NLINE.LT.KNL) GO TO 810	XEN	406
IF (NLINE.EQ.KNL) GO TO 820	XEN	407
NLINE=1	XEN	408
GO TO 900	XEN	409
810 NLINE=NLINE+1	XEN	410
GO TO 900	XEN	411
820 PRINT 830	XEN	412
830 FORMAT(1H1)	XEN	413
NLINE=0	XEN	414
GO TO 900	XEN	415
840 IF (NLINE.GE.KNL) GO TO 850	XEN	416
NLINE=NLINE+1	XEN	417
GO TO 880	XEN	418
850 PRINT 860,JW,A,NA,(Q(I),I=1,KTR)	XEN	419
860 FORMAT('1□□JW=',I3,4X,'A=',F8.2,4X,'NA=',I6,4X,'Q(I)=' ,12(F4.2,'/'	XEN	420
1),//)	XEN	421
PRINT 870	XEN	422
870 FORMAT(6X,'N',8X,'START',5X,'CLASS',4X,'INSTRM',4X,'PITCH',6X,	XEN	423
1'GLISS1',4X,'GLISS2',4X,'GLISS3',8X,'DURATION',5X,'DYNAM')	XEN	424
NLINE=1	XEN	425
880 PRINT 890,N,TA,KR,INSTRM,HX,(VIGL(I),I=1,3),XDUR,IFORM	XEN	426
890 FORMAT(1H ,I7,F12.2,I9,I8,F11.1,F13.1,2F10.1,F14.2,I11)	XEN	427
C	XEN	428
C PART 10,REPEAT SAME DEFINITIONS FOR ALL POINTS OF A	XEN	429
C	XEN	430
900 IF (N.LT.NA) GO TO 400	XEN	431
C	XEN	432
C PART 11, REPEAT SEQUENCES A	XEN	433
C	XEN	434
IF (KTEST2.EQ.0) GO TO 910	XEN	435
TAP2=TIMEF(1)-TAV2	XEN	436
TAP2=TAP2/FLOATF(NA)	XEN	437
PRINT 750,TAP2	XEN	438
C	XEN	439
910 IF (JW.GE.KW) GO TO 930	XEN	440
920 JW=JW+1	XEN	441
IF (GTNS.GT.SINA) GO TO 220	XEN	442
930 IF (KTEST1.EQ.0) CALL EXIT	XEN	443
940 TAP1=TIMEF(-1)-TAV1	XEN	444
TAP1=TAP1/FLOATF(KW)	XEN	445
PRINT 750,TAP1	XEN	446
C	XEN	447

END

XEN 448

C

C DATA FOR ATREES (ST/10-3, 060962)

C

000000011300022600033900045100056400067600078900090100101300112500123600
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0000336310999
00001953108000000101307200
00003487155000000157215500
25080408011309
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03030420010110
02050325010112
03350315011505
02100302103907
02020203150207
02020202410207
03090317041609
03132003200509
02052801030409
45011202020106

JW= 1 A= 9.13 NA= 95
Q(1)=0.12/0.04/0.04/0.05/0.12/0.29/0.04/0.04/0.14/0.06/0.06/0.03/

N	START	CLASS	INSTRM	PITCH	GLISS1	GLISS2	GLISS3	DURATION	DYNAM
1	0.00	7	1	34.0	0.0	0.0	0.0	0.00	3
2	0.10	10	1	43.2	0.0	0.0	0.0	0.41	50
3	0.11	6	8	81.3	0.0	0.0	0.0	0.63	21
4	0.13	6	3	47.0	0.0	0.0	0.0	0.18	10
5	0.18	1	4	0.0	0.0	0.0	0.0	1.90	29
6	0.25	9	1	48.7	0.0	0.0	0.0	1.51	35
7	0.33	6	7	11.4	0.0	0.0	0.0	0.37	42
8	0.34	9	1	38.1	0.0	0.0	0.0	0.00	59
9	0.40	1	1	0.0	0.0	0.0	0.0	2.20	45
10	0.41	6	9	55.0	0.0	0.0	0.0	1.07	0
11	0.76	6	7	11.5	0.0	0.0	0.0	0.40	7
12	0.90	8	2	23.2	0.0	0.0	0.0	0.00	19
13	1.00	7	2	26.9	0.0	0.0	0.0	0.00	6
14	1.09	10	1	46.2	0.0	0.0	0.0	0.32	57
15	1.09	6	2	68.5	0.0	0.0	0.0	0.71	25
16	1.23	6	3	46.9	0.0	0.0	0.0	0.64	32
17	1.42	6	1	44.0	0.0	0.0	0.0	0.44	1
18	1.57	10	1	36.2	0.0	0.0	0.0	0.22	21
19	1.65	4	2	32.5	0.0	0.0	0.0	1.09	13
20	1.78	6	8	72.6	0.0	0.0	0.0	0.06	60
21	1.92	6	3	38.9	0.0	0.0	0.0	0.55	60
22	1.94	5	1	74.6	71.0	-25.0	-71.0	0.80	62
23	2.18	4	1	32.6	0.0	0.0	0.0	1.50	50
24	2.18	6	6	50.9	0.0	0.0	0.0	0.60	26
25	2.19	1	12	0.0	0.0	0.0	0.0	4.58	24
26	2.20	9	1	49.3	0.0	0.0	0.0	0.02	58
27	2.23	9	1	51.0	0.0	0.0	0.0	0.22	13
28	2.32	7	1	36.9	0.0	0.0	0.0	0.00	43
29	2.33	4	1	31.8	0.0	0.0	0.0	1.38	56
30	2.54	1	6	0.0	0.0	0.0	0.0	0.28	14
31	2.57	11	2	12.2	0.0	0.0	0.0	1.69	40
32	2.71	9	1	48.5	0.0	0.0	0.0	0.37	55
33	2.80	1	5	0.0	0.0	0.0	0.0	1.50	58
34	3.28	5	2	15.4	49.0	5.0	-31.0	0.52	21
35	3.33	1	7	0.0	0.0	0.0	0.0	1.38	8
36	3.38	5	2	47.3	-71.0	-17.0	46.0	1.05	4
37	3.55	10	1	37.6	0.0	0.0	0.0	0.14	24
38	3.56	1	9	0.0	0.0	0.0	0.0	1.30	0
39	3.60	9	1	64.3	0.0	0.0	0.0	0.19	12
40	3.64	12	2	52.2	0.0	0.0	0.0	3.72	9
41	3.65	6	5	59.0	0.0	0.0	0.0	0.83	28
42	3.71	5	3	38.8	25.0	2.0	-15.0	0.00	11
43	3.80	6	8	75.6	0.0	0.0	0.0	0.43	17
44	3.87	6	2	51.5	0.0	0.0	0.0	0.77	57
45	3.89	6	7	12.1	0.0	0.0	0.0	0.39	2
46	4.15	5	2	43.0	-71.0	24.0	71.0	1.16	2
47	4.15	5	1	80.3	36.0	4.0	22.0	0.85	50
48	4.25	9	1	59.9	0.0	0.0	0.0	0.10	10
49	4.31	12	2	40.1	0.0	0.0	0.0	2.49	33
50	4.33	1	10	0.0	0.0	0.0	0.0	0.46	34

Appendix I

Two Laws of Continuous Probability

First Law

$$P_x = ce^{-cx} dx.$$

Let OA be a segment of a straight line of length ℓ on which we place n points. Their linear density is $c = n/\ell$. Suppose that ℓ and n increase indefinitely while c remains constant. Suppose also that these points are numbered $A_1, A_p, A_q, \dots$ and are distributed from left to right beginning at the origin 0. Let

$$x_1 = A_1A_p, \quad x_2 = A_pA_q, \quad x_3 = A_qA_r, \dots, x_i = A_sA_t.$$

The probability that the i -th segment will have a length x_i between x and $x + dx$ is

$$P_x = ce^{-cx} dx.$$

Now the probability p_n , that there will be n points on a segment x is given by the recurrence formula

$$\frac{p_{n+1}}{p_n} = \frac{cx}{n+1}.$$

Therefore $p_1 = (cx/1)p_0$. But $p_0 = e^{-cx}$ and

$$e^{-cx} = 1 - \frac{cx}{1!} + \frac{(cx)^2}{2!} - \frac{(cx)^3}{3!} + \dots.$$

If x is very small and if we denote it by dx , we have

$$p_0 = 1 - cdx + \frac{c^2(dx)^2}{2!} - \dots.$$

Since the powers of dx are infinitely small for high values, $p_0 = 1 - cdx$ and $p_1 = cdxp_0 = cdx$. Hence, the probability P_x is composed of the probability $p_0 = e^{-cx}$, that there will be no point on the segment x , and the probability $p_1 = cdx$, that there will be a point in dx .

Approximate Calculation of the Same Probability (For Calculation by Hand)

Let there be d points to be placed on a straight line of length ℓ . The linear density is $c = d/\ell$ points on length ℓ . If the lengths are expressed in units v then $\ell = av$ ($a > 0$) and $cv = d/a$ points in the unit of length v .

Then $x_i = iv$ ($i = 0, 1, 2, 3, \dots$), and the probability, the asymptotic limit of the relative frequency of the segment x_i , will be

$$P_{x_i} = ce^{-civ} \Delta x_i. \quad (1)$$

We shall now define the quantity Δx_i . The probability (1) is composed of the probability $p_0 = e^{-civ}$, that there will be no point on x_i , and the probability $p_1 = c\Delta x_i$ that there will be a point in Δx_i if $(c\Delta x_i)^2$ is small enough to be ignored. Set

$$0 < (c\Delta x_i)^2 < w10^{-n},$$

where n is a sufficiently large natural number; this expression becomes

$$0 < \Delta x_i < c^{-1} \cdot 10^{-n/2}.$$

Substitute a constant z for Δx_i such that for every x_i

$$z \leq \Delta x_i < c^{-1} \cdot 10^{-n/2}. \quad (2)$$

Then equation (1) is written

$$P_{x_i} = e^{-civ} \cdot cz \quad (3)$$

and must satisfy the condition

$$\sum_{i=0}^{\infty} cze^{-civ} = 1,$$

or

$$z = \frac{1}{c} \sum_{i=0}^{\infty} e^{-civ}.$$

But since $cv > 0$, $e^{-cv} < 1$, so that

$$\sum_{i=0}^{\infty} (e^{-cv})^i = \frac{1}{1 - e^{-cv}}$$

and finally

$$z = \frac{1 - e^{-cv}}{c}.$$

Now from 2

$$\frac{1 - e^{-cv}}{c} < \frac{10^{-n/2}}{c}.$$

Therefore

$$0 < (1 - e^{-cv}) < 10^{-n/2},$$

then

$$(1 - 10^{-n/2}) < e^{-cv} < 1$$

Thus, for $cv > 0$ we have $e^{-cv} < 1$, and for $cv < -\log(1 - 10^{-n/2})$ we have

$$-\log(1 - 10^{-n/2}) = 10^{-n/2} + \frac{10^{-(n/2)^2}}{2} + \frac{10^{-(n/2)^3}}{3} + \frac{10^{-(n/2)^4}}{4} + \dots$$

and

$$10^{-n/2} < -\log(1 - 10^{-n/2}).$$

In order that $e^{-cv} > 1 - 10^{-n/2}$ it is therefore sufficient that

$$cv \leq 10^{-n/2}. \quad (4)$$

Then we may take

$$\Delta x_i = z = \frac{1 - e^{-cv}}{c} \quad (5)$$

and substitute this value in formula (1), from which we can now set up probability tables. Here is an example :

Let $d = 10$ points as mean value to be spread on a straight line segment of length $\ell = 100$ cm. We have to define x_i and P_{x_i} as a function of i , given that $(c\Delta x_i)^2 = 10^{-4}$ is considered to be negligible.

From (4), $cv = 10^{-4/2} = 0.01$ points in v . Now $c = d/\ell = 10/100\text{points/cm}$, therefore $c = 0.1$ points/cm, $v = 0.01/0.1 = 0.1$ cm, and $x_i = 0.1 \cdot icm = imm$. From (5),

$$\Delta x_i = \frac{1 - e^{-0.01}}{0.1} = (1 - 0.9905)10 = 0.0995 = 0.1\text{cm}.$$

From (1),

$$P_{x_i} = e^{-0.01i} \cdot 0.1 \cdot 0.1 = 0.01 \cdot (0.99005)^i.$$

For calculation by machine see Chapter V.

Second Law

$$f(j) dj = \frac{2}{a} \left(1 - \frac{j}{a}\right) dj.$$

Each variable (pitch, intensity, density, etc.) forms an interval (distance) with its predecessor. Each interval is identified with a segment x taken on the axis of the variable. Let there be two points A and B on this axis corresponding to the lower and upper limits of the variable. It is then a matter of drawing at random a segment within AB whose length is included between j and $j + dj$ for $0 \leq j \leq AB$. Then the probability of this event is:

$$P_j = f(j) dj = \frac{2}{a} \left(1 - \frac{j}{a}\right) dj \quad (1)$$

for $a = AB$.

Approximate Definition of this Probability for Calculation by Hand

By taking dj as a constant and j as discontinuous we set $dj = c$, $j = iv$ with $v = a/m$ for $i = 0, 1, 2, 3, \dots, m$. Equation (1) becomes

$$P_j = \frac{2}{a} \left(1 - \frac{iv}{a}\right) c. \quad (2)$$

But

$$\sum_{i=0}^m P_j = \frac{2c}{a}(m+1) - \frac{2cv}{a^2} \sum_{i=0}^m i = \frac{2c(m+1)}{a} - \frac{2cvm(m+1)}{2a^2} = 1,$$

whence

$$dj = c = \frac{a}{m+1}.$$

On the other hand P_j must be taken as a function of the decimal approximation required:

$$P_j = \frac{2}{m+1} \left(1 - \frac{i}{m}\right) \quad (n = 0, 1, 2, 3, \dots).$$

P_j is at a maximum when $i = 0$, whence $m \geq 2 \cdot 10^n - 1$; so for $m = 2 \cdot 10^n - 1$ we have $v = a/(2 \cdot 10^n - 1)$ and $dj = a/(2 \cdot 10^n)$, and (1) becomes

$$P_j = P_i = \frac{1}{10^n} \left(1 - \frac{i}{2 \cdot 10^n - 1}\right).$$

Definition of the Same Probability for Computer Calculation

We know that the computer can only draw numbers y_0 at random (of equal probability) $0 \leq y_0 \leq 1$. Using the probability law of density $P_j = f(j) \, dj$, we have for some interval x_0

$$\text{prob. } (0 \leq j \leq x_0) = \int_0^{x_0} f(j) \, dj = \frac{2x_0}{a} - \frac{x_0^2}{a^2} = F(x_0),$$

where $F(x_0)$ is the distribution function of j . But $F(x_0) = \text{prob. } (0 \leq y \leq y_0) = y_0$. Therefore

$$\frac{2x_0}{a} - \frac{x_0^2}{a^2} = y_0 \quad \text{and} \quad x_0 = a \left(1 \pm \sqrt{1 - y_0} \right),$$

and by rejecting the positive root, since x_0 must remain smaller than a , we obtain

$$x_0 = a \left(1 - \sqrt{1 - y_0} \right)$$

for all $0 \leq x_0 \leq a$.

Table 1: Composition of the Orchestra for *Atrées* (ST/10-3,060962)

Timbre classes and instruments as on present input data

<i>Class</i>	<i>Timbre</i>	<i>Instrument</i>	<i>Instrument No.</i>
1	Percussion	Temple-blocks	1 — 5
		Tom-toms	6 — 9
		Maracas	10
		Susp. Cymbal	11
		Gong	12
2	Horn	French horn	1
3	Flute	Flute	1
4	Clarinet	Clarinet B	1
		Bass clar. B	2
5	Glissando	Violin	1
		Cello	2
		Trombone	3
6	Tremolo or flutter-tongue	Flute	1
		Clarinet Bb	2
		Bass clar. Bb	3
		French horn	4
		Trumpet	5
		Trombone a	6
		Trombone b	7
		(pedal notes)	
		Violin	8
		Cello	9
7	Plucked strings	Violin	1
		Cello	2
8	Struck strings	Violin	1
		Cello	2
9	Vibraphone	Vibraphone	1
10	Trumpet	Trumpet	1
11	Trombone	Trombone a	1
		Trombone b	2
12	Bowed strings	(pedal notes)	
		Violin	1
		Cello	2